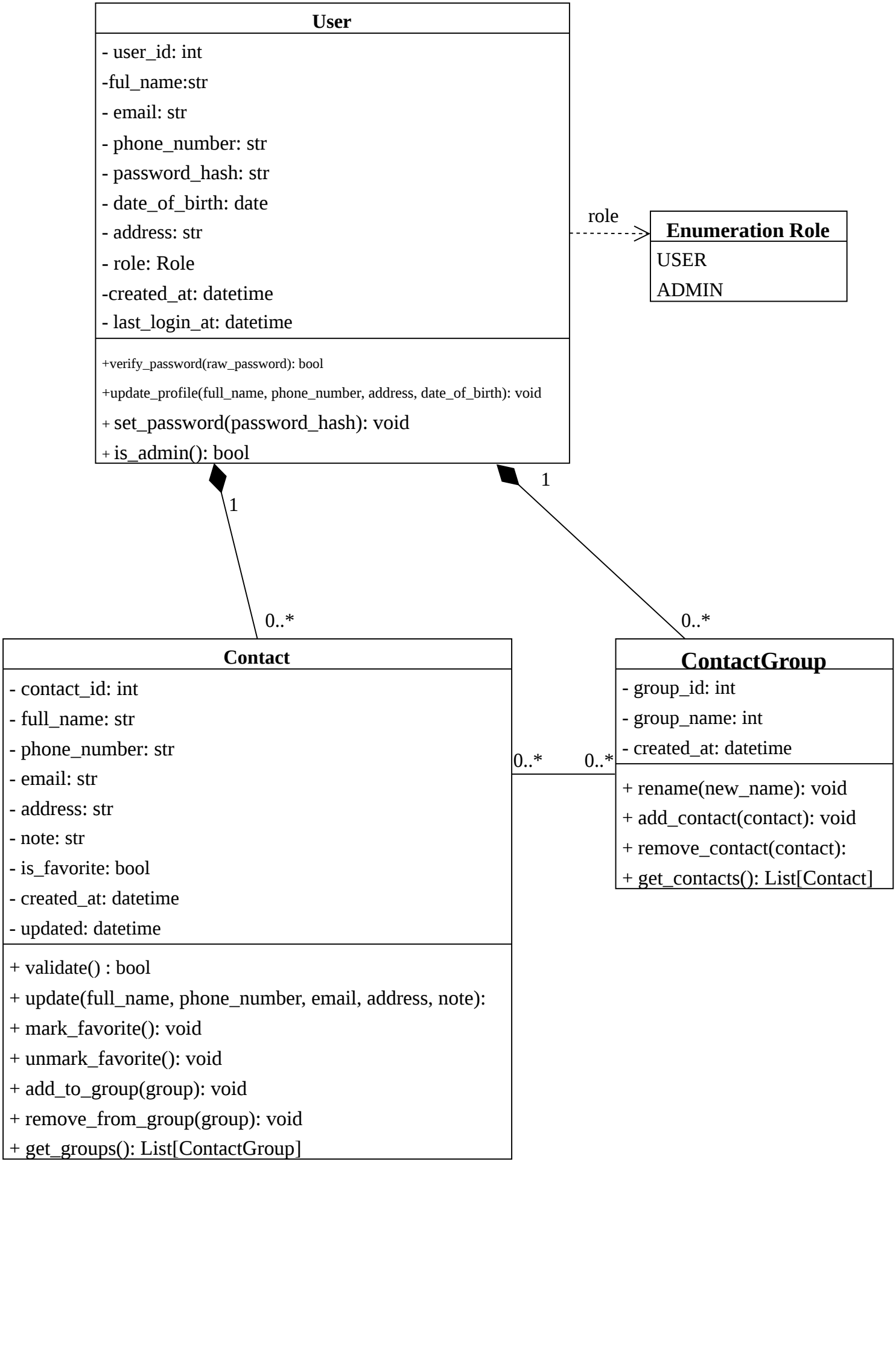
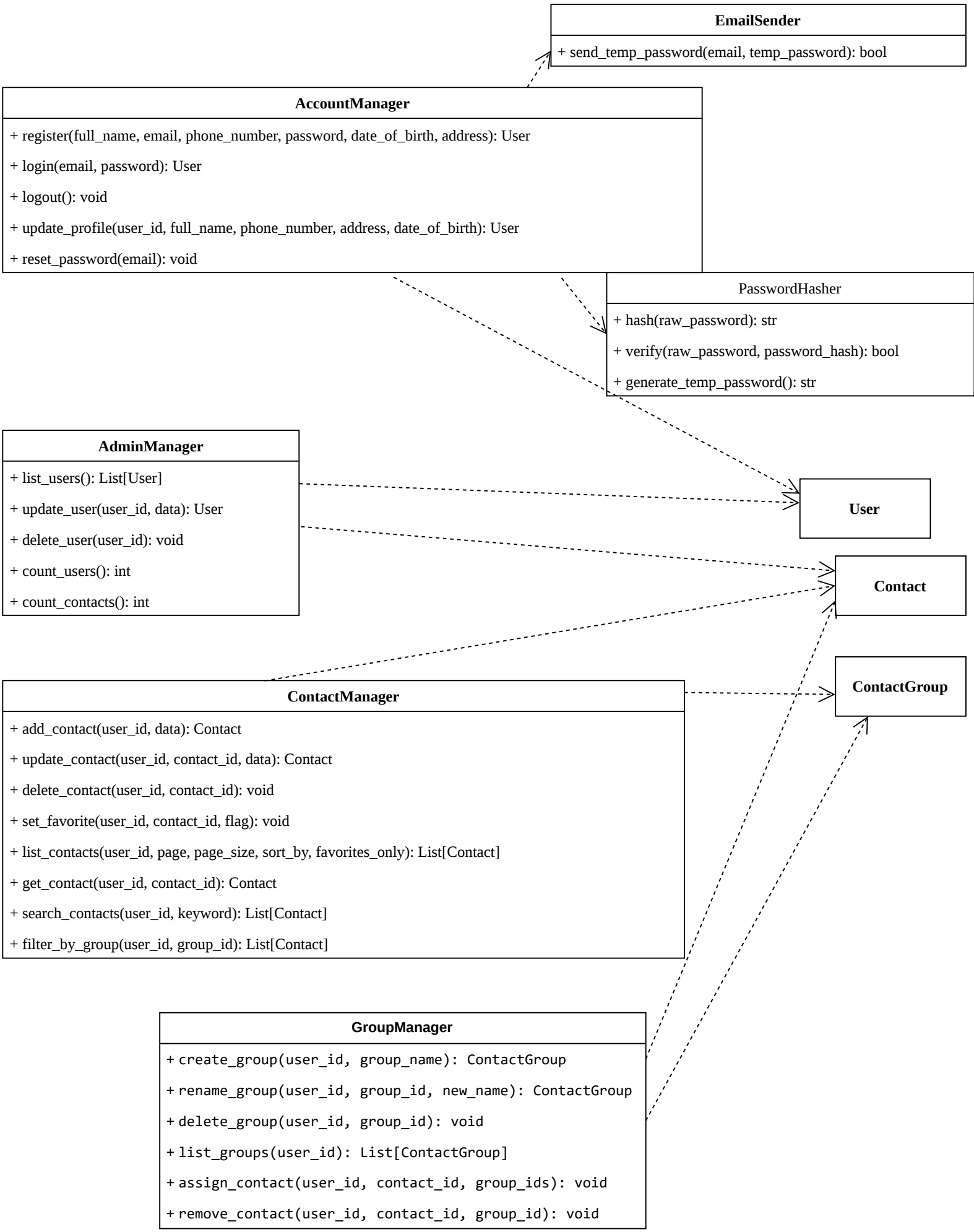
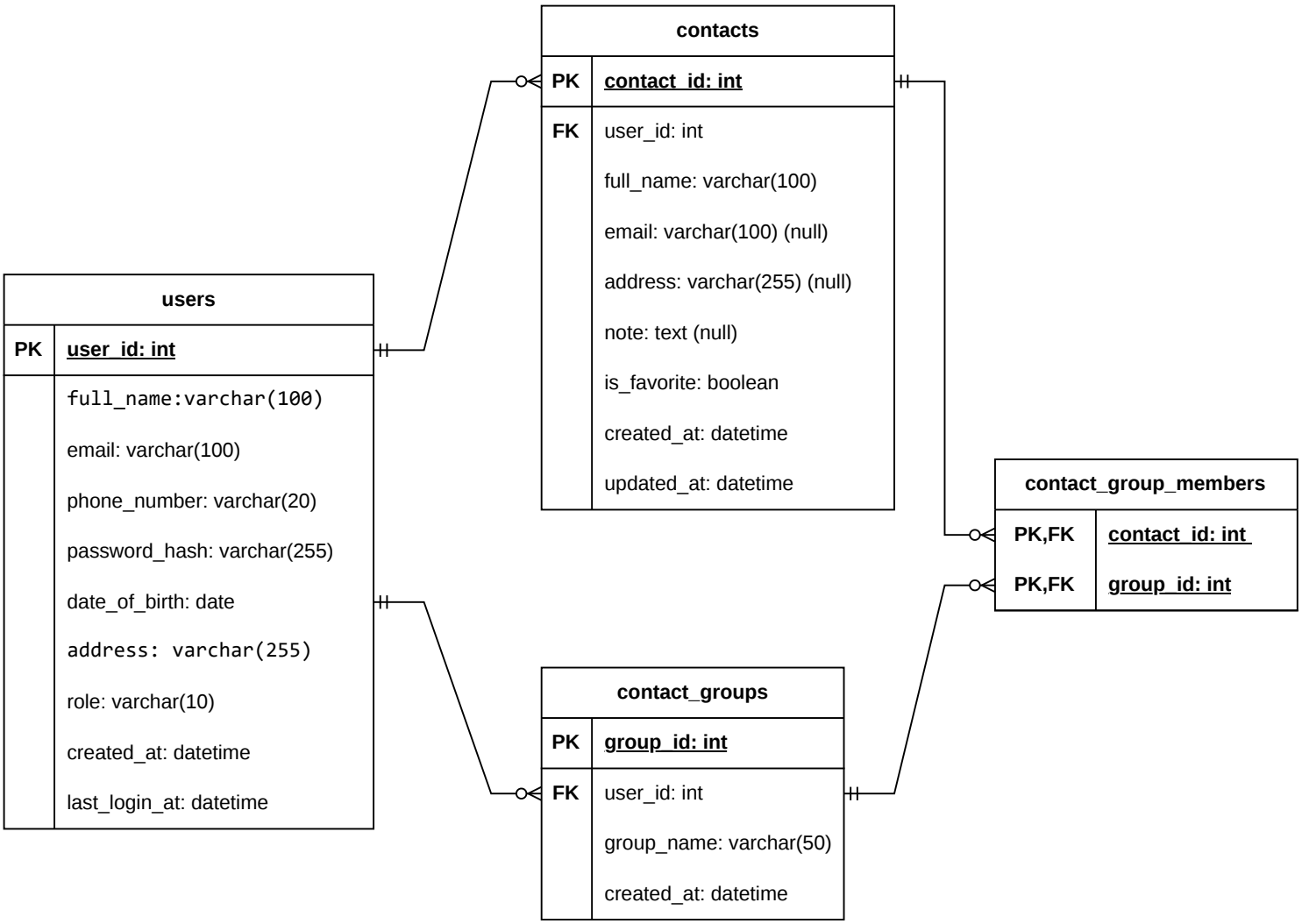

User
- user_id: int
-ful_name:str
- email: str
- phone_number: str
- password_hash: str
- date_of_birth: date
- address: str
- role: Role
-created_at: datetime
- last_login_at: datetime
+verify_password(raw_password): bool
+update_profile(full_name, phone_number, address, date_of_birth): void
+ set_password(password_hash): void
+ is_admin(): bool
role
Enumeration Role
USER
ADMIN
1
0..*
1
0..*
Contact
- contact_id: int
- full_name: str
- phone_number: str
- email: str
- address: str
- note: str
- is_favorite: bool
- created_at: datetime
- updated: datetime
+ validate() : bool
+ update(full_name, phone_number, email, address, note):
+ mark_favorite(): void
+ unmark_favorite(): void
+ add_to_group(group): void
+ remove_from_group(group): void
+ get_groups(): List[ContactGroup]
0..*
0..*
ContactGroup
- group_id: int
- group_name: int
- created_at: datetime
+ rename(new_name): void
+ add_contact(contact): void
+ remove_contact(contact):
+ get_contacts(): List[Contact]


EmailSender
+ send_temp_password(email, temp_password): bool
AccountManager
+ register(full_name, email, phone_number, password, date_of_birth, address): User
+ login(email, password): User
+ logout(): void
+ update_profile(user_id, full_name, phone_number, address, date_of_birth): User
+ reset_password(email): void
PasswordHasher
+ hash(raw_password): str
+ verify(raw_password, password_hash): bool
+ generate_temp_password(): str
AdminManager
+ list_users(): List[User]
+ update_user(user_id, data): User
+ delete_user(user_id): void
+ count_users(): int
+ count_contacts(): int
User
Contact
ContactGroup
ContactManager
+ add_contact(user_id, data): Contact
+ update_contact(user_id, contact_id, data): Contact
+ delete_contact(user_id, contact_id): void
+ set_favorite(user_id, contact_id, flag): void
+ list_contacts(user_id, page, page_size, sort_by, favorites_only): List[Contact]
+ get_contact(user_id, contact_id): Contact
+ search_contacts(user_id, keyword): List[Contact]
+ filter_by_group(user_id, group_id): List[Contact]
GroupManager
+ create_group(user_id, group_name): ContactGroup
+ rename_group(user_id, group_id, new_name): ContactGroup
+ delete_group(user_id, group_id): void
+ list_groups(user_id): List[ContactGroup]
+ assign_contact(user_id, contact_id, group_ids): void
+ remove_contact(user_id, contact_id, group_id): void


contacts
PK
contact_id: int
FK
user_id: int
full_name: varchar(100)
email: varchar(100) (null)
address: varchar(255) (null)
note: text (null)
is_favorite: boolean
created_at: datetime
updated_at: datetime
users
PK
user_id: int
full_name:varchar(100)
email: varchar(100)
phone_number: varchar(20)
password_hash: varchar(255)
date_of_birth: date
address: varchar(255)
role: varchar(10)
created_at: datetime
last_login_at: datetime
contact_group_members
PK,FK
contact_id: int
PK,FK
group_id: int
contact_groups
PK
group_id: int
FK
user_id: int
group_name: varchar(50)
created_at: datetime